Smart Energy Research Lab: Energy Use in GB domestic buildings 2021

Aggregated statistics documentation

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Table of Contents

Introduction

The vision of the Smart Energy Research Lab (SERL) is to provide a secure, consistent, and trusted channel for researchers to access high-resolution energy data, which will facilitate innovative energy research for years to come. SERL is funded by the UK's Engineering and Physical Sciences Research Council (EPSRC) and comprises a consortium of seven UK universities and the Energy Saving Trust. We have established an Observatory panel of ~13,000 smart metered households across Great Britain, and we provide high-resolution Observatory data to UK researchers via the UK Data Service (UKDS).

This document describes a dataset of aggregated energy consumption statistics from the SERL 4th Edition for 2021 and 2020. These aggregated statistics are also available via the UKDS to all registered/authenticated users.

The aggregated statistics are provided as a single long-form csv file containing 109,990 records:

- *serl_energy_use_in_GB_domestic_buildings_2021_aggregated_statistics.csv*

This document is structured as follows: the method by which the aggregated statistics were produced is described, followed by data dictionaries explaining the fields within the aggregated statistics in detail.

We will make the code used to generate these statistics available via the SERL GitHub repository. We encourage data users to provide feedback, requests for improvements or future additional aggregated statistics (contact info@serl.ac.uk).

A subset of these statistics were used to generate the results presented in the **'Smart Energy Research Lab: Energy use in GB domestic buildings 2021' report**, available via the SERL website.

There is extensive documentation covering the underlying controlled SERL Observatory dataset upon which this aggregated statistical dataset was based, available via UKDS (SN 8666).

Generation of aggregated statistics

All the outputs described in this report are based on statistical analysis of the SERL Observatory dataset SN 8666:

Elam, S., Webborn, E., McKenna, E., Oreszczyn, T., Anderson, B., Few, J., Pullinger, M., European Centre for Medium-Range Weather Forecasts, Ministry of Housing, Communities and Local Government, Royal Mail Group Limited, 2022, *Smart Energy Research Lab Observatory Data, 2019-2021: Secure Access*, [data collection], UK Data Service, 5th Edition. SN: 8666, DOI: 10.5255/UKDA-SN-8666-5

The statistical dataset described here consists of aggregated summaries of the net electricity and gas consumption data with aggregated linked contextual data, optionally grouped by one of the categorical variables (e.g. by region, by EPC rating) or by banded continuous variables (e.g. temperature band) available in the contextual data.

The summaries are provided at the following levels of aggregation:

1. Annual – annual averages of daily net electricity or gas consumption, external temperature, and heating degree days
2. Monthly – monthly averages of daily net electricity or gas consumption, external temperature, and heating degree days
3. Diurnal – annual averages of half-hourly net electricity or gas consumption, and hourly external temperature, all reported in local time

The categorical variables which are used for grouping are:

- Temperature – data is grouped by mean external temperature bands:
  - 0 to 5 °C, 5 to 10 °C, 10 to 15 °C, 15 to 20 °C
- Region:
  - Former Government Office Regions for England plus Wales and Scotland
- Index of Multiple Deprivation (IMD) 2019 quintile (1 to 5) for the Lower Layer Super Output Area (LSOA) in which the dwelling is located
- Energy Performance Certificate rating – from A to G
- Floor area of building (in bands)
- Heating system e.g. gas boiler, electric radiators, etc.
- Year of construction of building (in bands)
- Building type

- Number of occupants
- Tenure
- Electric vehicle ownership
- Photovoltaic ownership

Note on groupings: the number of observations in each sub-group was manually checked and where less than 10 the sub-group was merged with another sub-group to prevent disclosure. For example, EPC bands A and B were grouped together. The full list of merged sub-groups can be inspected in the python code released via the SERL github repository. Also note that the sub-groups contain different participants for electricity and gas statistics.

Two sets of households are used in these summaries:

1. All households with sufficient data to estimate an annual average for the year 2021
2. All households with sufficient data to estimate annual averages for both the year 2020 and 2021

As recruitment of participants to the SERL project occurred in waves, the first set has more households and includes the households that are in the second set. As it is larger, we use the first set for all outputs relating to 2021. However, a small number of outputs use the second, smaller set to provide information about how energy consumption changed from 2020 to 2021 *for the same households over time*. Indicative sizes of the two sets: first set has >10k households for electricity and ~8k for gas, while the second set has ~3k for electricity and for gas. Note that the actual N for each statistic may be substantially lower than this due to contextual variable segmentation and missing data.

The process for producing these is as follows.

1. Household-level aggregation

For each household, valid[1] half-hourly and daily electricity and gas consumption, and hourly external temperature data were aggregated for each month to produce:

- average (mean) and percentiles of half-hourly consumption and mean external temperature for each month and
- average (mean) and percentiles of daily consumption and mean daily external temperature, and mean heating degree days for each month.

For each household, these monthly aggregations were then aggregated over the whole year to produce an average half-hourly profile (mean for each half-hour over all months) for each year and to produce average daily values (mean daily consumption, mean temperature and mean heating degree days over all months) for each year.

Note on data quality thresholds: We impose two data quality thresholds:

1. We require **at least 50% valid half-hourly reads** to be present to estimate the average for a household for any given month. For example, at least 50% of the half-hourly reads in July at 13:30 must be valid to estimate the average 13:30 July energy use for each fuel type.
2. We also require **all** monthly aggregations to be present to produce the annual aggregations (i.e. 0% missing data threshold). Any household with missing months is excluded. This ensures that if data is missing in any particular season, this will not skew the annual result..

This results in

- **household-level annual summaries of daily averages**
- **household-level monthly summaries of daily averages**
- **household-level annual summaries of half-hourly averages**

1 Data that were valid and on time according to standard SERL data processing procedures, for example this identifies outliers and issues with units, see our Smart Meter Documentation for more information here.

This step is repeated with household-level data grouped by temperature band (e.g. household-level averages of daily consumption on days with mean temperature between 0-5ºC) and for weekdays vs weekends. This results in a further set of household-level summaries *for each temperature band and for weekends/weekdays*.

Note the household-level annual summaries are **not** included in this statistical dataset but are used to create the outputs in this dataset and the 'grouping-level aggregated outputs' described below.

2. Grouping-level aggregation

Groupings of households into the categories listed above are used for specific outputs. This could be 'all households' or could be determined by a categorical variable segmentation such as 'all households in Region equals Scotland'.

For each grouping, the following household-level summaries are calculated for the households that are members of that grouping:

- Mean of:
  - Electricity and gas consumption
  - External temperature
  - Heating degree days (for daily summaries only)
- Standard deviation of:
  - Electricity and gas consumption
- Percentiles (25^{th}, 50^{th}, 75^{th}) of:
  - Electricity and gas consumption
- N – the number of household-level summaries used to compute the statistic

Note on percentiles: the N referenced for each percentile calculated refers to the number of observations relating to the underlying percentile value. We ensure percentiles refer to at least 10 observations by reporting the mean of the ten closest observations to the percentile; this is to minimize the risk of statistical disclosure of participant identities.

This results in:

- **Grouping-level annual summaries of daily averages**
- **Grouping-level monthly summaries of daily averages**
- **Grouping-level annual summaries of half-hourly averages**

These grouping level summaries are then used to create the following output types:

- Descriptive statistics – mean, median and percentiles (25^{th}, median, 75^{th})

For each grouping we also report the following key summary statistics for the SERL Observatory as a whole (rather than just the subset which had sufficient energy data to contribute to the given statistic):

- Mean floor area (only available for homes with an EPC – about half of the sample)
- Mean number of occupants
- Mean number of bedrooms

For example, we calculate these averages for all households in the SERL Observatory with region equals Scotland, and so on for all other groupings. Alongside this we report the number of cases used to calculate these means.

We will make the code used to produce the statistical dataset and outputs available on the SERL GitHub repository.

Data Dictionaries

Table 1 lists the variables in the aggregated statistics dataset with a short description of the information provided by this variable. Table 2 provides further details on the categories within the aggregated dataset, and Table 3 provides the segments within each segmentation variable. Note that the detailed coding of these categories can be inspected via the Python code available on the SERL GitHub repository.

Table 1. Guide to the aggregated statistics variables

Variable	*Class*	*N unique values*	*Description*	*Example value*
fuel	String	2	Fuel that the value represents	Net electricity
unit	String	2	Unit of the value	Wh
summary_stat	String	5	What statistical summary the value represents	Median
subsample	String	2	What subsample of SERL Observatory homes were used	All
summary_time	String	76	Time period that the value relates to, either a specific half hour or a daily value. Note that for half hourly statistics, the reported time is local time, and the value is for the energy used in the preceding 30 minutes.	00:30
time_period	String	6	Time period that the data is summarized over	2021
segmentation_variable_1	String	17	Variable used to segment the Observatory sample	num_occupants
segment_1_value	String	75	Segment of the above segmentation variable that this value relates to (any value that the segmentation variable can take, or a merge of 2+ values to ensure at least 10 cases are included)	>=5

Variable	*Class*	*N unique values*	*Description*	*Example value*
value	Float	85,559	Value of the specified summary statistic, summary period and segment	8.168
n_sample	Float	3,319	Number of cases in the distribution this value was drawn from	10,764
n_statistic	Float	3,320	Number of cases used to calculate the value. For mean and standard deviation n_sample = n_statistic. For centiles n_statistic = 10 as we take the 10 closest values to the centile and report the mean of these in the value field for SDC reasons.	10
mean_temp	Float	6869	Mean external temperature in ^{o}C over the summary time during the time period	10.378
mean_hdd	Float	2613	Mean heating degree days over the summary time during the time period. Note that this is NaN for half hourly summary times as heating degree days are defined for daily periods.	5.74
decimal_places	Float	1	Number of decimal places the value has been rounded to	3
weekday_weekend	String	4	Whether the value was calculated using only data from weekdays, weekends or both	Weekday
mean_floor_area	Float	77	Mean floor area in m^2 for dwellings in the SERL observatory with segment_1_value and a floor area value from an EPC	100.026

Variable	*Class*	*N unique values*	*Description*	*Example value*
n_mean_floor_area	Float	76	Number of cases used to calculate mean_floor_area	1437
mean_bedrooms	Float		Mean number of bedrooms for dwellings in the SERL observatory with segment_1_value and with an answer to the number of bedrooms question (B6) in the SERL survey	2.918
n_mean_bedrooms	Float		Number of cases used to calculate mean_bedrooms	2586
mean_occupants	Float		Mean number of occupants for dwellings in the SERL observatory with segment_1_value and with a valid answer to the number of occupants question (C1_new) in the SERL survey	2.297
n_mean_occupants	Float		Number of cases used to calculate n_mean_occupants	2581

Table 2. Guide to categories within the aggregated dataset

Variable	Value	Description
Subsample	All	All cases with sufficient data within the specified summary time, time period and segment value
Subsample	participants with annual data for 2021 and 2020	Cases with sufficient daily data in each month of 2020 and 2021
summary_time	heating_season_yyyy_yyyy	We take the SAP heating season of October to May inclusive
time_period	heating_season_yyyy_yyyy	We take the SAP heating season of October to May inclusive
segmentation_variable_1	num_occupants	Values from SERL survey variable C1_new
segmentation_variable_1	num_bedrooms	Values from SERL survey variable B6
segmentation_variable_1	building_type	Values from SERL survey variable B1
segmentation_variable_1	building_age_merge	Values from SERL survey variable B9
segmentation_variable_1	tenure	Values from SERL survey variable B4

Variable	*Value*	*Description*
segmentation_variable_1	boiler_type_merge_for_elec_consumption	Values from SERL survey question A3. Note that different groups are used for electricity and gas consumption statistics for the boiler type variable because homes with electric heating rarely also have a gas supply. If the same groupings were used many categories would have to be merged to ensure cases >10 for all statistics.
segmentation_variable_1	boiler_type_merge_for_gas_consumption	Values from SERL survey question A3. Note that different groups are used for electricity and gas consumption statistics for the boiler type variable because homes with electric heating rarely also have a gas supply. If the same groupings were used many categories would have to be merged to ensure cases >10 for all statistics.
segmentation_variable_1	has_ev_merge	Values from SERL survey variable C5
segmentation_variable_1	currentEnergyRating_merge	Values from EPC variable currentEnergyRating
segmentation_variable_1	floor_area_banded	Values from EPC variable totalFloorArea
segmentation_variable_1	has_PV	Participants are included in this group if they have any positive readings in the half-hourly electricity export column for the given year, or the EPC variable photoSupply is more than 0.

Table 3. Detailed coding of segment values

Segmentation_variable_1	*Possible segment values*
boiler_type_merge_for_elec_consumption	Gas boiler, Gas boiler plus other, Electric storage radiators, Electric radiators, Other electric, District or community, Oil, solid fuel or biomass, Other or other mix, None
boiler_type_merge_for_gas_consumption	Gas boiler, Gas boiler plus other, Not gas
building_age_merge	Before 1900, 1900 - 1929, 1930 - 1949, 1950 - 1975, 1976 - 1990, 1990 - 2002, 2003 onwards, No data
building_type_merge	Detached, Semi-detached, Terraced, Purpose-built flat, Converted flat or shared house, Other or no answer
currentEnergyRating_merge	A and B, C, D, E, F and G
floor_area_banded	50 or less, 50 to 100, 101 to 150, 151 to 200, Over 200
has_ev_merge	Yes, No, No data
has_pv	TRUE, FALSE
IMD_quintile	1, 2, 3, 4, 5
num_bedrooms	1, 2, 3, 4, >=5, No data
num_occupants	1, 2, 3, 4, 5, >=6, No data

Segmentation_variable_1	*Possible segment values*
Region	South West, South East, Wales, Greater London, East of England, East Midlands, West Midlands, North West, Yorkshire, North East, Scotland
temperature_band	0 to 5, 5 to 10, 10 to 15, 15 to 20, 4.5 to 5.5
tenure	Own outright or mortgage, Part-own part-rent, Private rent, Social rent, Rent free, No answer
weekday_weekend	Weekday, weekend